

Seb Blanchet

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Skills

Languages : Python, Rust, Bash, C, C++, C#, JavaScript, HTML, Lua, SQL, Elixir, Swift
Tools : Git, Wireshark, Neovim, Jenkins, Docker, K8, AWS, LaTeX, Markdown, Nginx, Bazel, Nix
Technologies : Numpy, Torch, Tensorflow, Pandas, Vue, React, WASM, OpenCV, CUDA, ffmpeg, ollama
Platforms : Intel x86-64, Apple Silicon, ARM, Arduino, STM32, NXP, Raspberry Pi, Xilinx, NVIDIA GPU, RISC-V
Simulation : MATLAB, LabVIEW, Simulink, OPAL, Speedgoat, dSPACE, SOLIDWORKS, ANSYS
Os : Windows, macOS, iOS, Ubuntu, Arch, Debian, Asahi, Fedora, RHEL, Kali, FreeRTOS, QNX
Protocols : TLS, SSH, PGP, HTTP/2/3, TCP/IP, UDP, DNS, XCP, CAN, LIN, UDS, SPI, I2C, JTAG, UART
Concepts : AI/ML, LLM, MIMO, PID, DSP, HIL, TDD, OOP, DSA, CI/CD, REST, UNIX, OSI, OWASP

Education

University of Waterloo
Bachelor of Applied Science with Distinction
• Honors Mechanical Engineering Co-op - GPA: 3.5 / 4.0
Sep 2013 - Jun 2019 (5yr, 10mo)
Waterloo, ON, CAN

Experience

NVIDIA
Staff Software Engineer - LPX Firmware
• Developing low-level RISC-V C firmware for AI inference accelerator LPX compute tray
• Leading design and implementation of control-plane telemetry interfaces and host tooling in C++
Jan 2026 - Present (0yr, 2mo)
Remote

Groq
Staff Software Engineer - Devices Firmware
• Shipped production C/C++ firmware for hardware-accelerated AI/LLM inference across datacenter fleet
• Developed host-side tooling in Python/ Rust for communication and orchestration with RISC-V/ARM targets
• Implemented Linux kernel drivers and BSP integration within Yocto-based embedded Linux environments
• Supported silicon bring-up and validation for initial SoC delivery, including lab infrastructure setup
• Troubleshooted serial communication (I2C, SPI, CAN) hands-on and calibrating sensors with lab equipment
• Led a remote engineering team and coordinated project tasks to ensure efficient workflow
Dec 2024 - Dec 2025 (1yr, 0mo)
Remote

Apple
Senior Software Engineer - Human Interface Devices
• Automated iOS/macOS testing and deployment using Bash/ Python for all devices supporting Touch ID
• Collected and analyzing sensor data to extract core metrics and perform statistical analysis and visualizations
• Integrated custom Swift native apps into the testing workflows to interact with internal SDK APIs
• Streamlined AI/ML data processing, model training tasks with automation scripts and cloud computing
• Scaled and managing Kubernetes cloud infrastructure for Docker and Rust based web apps and services
May 2024 - Nov 2024 (0yr, 7mo)
Remote

Apple
Senior Software Engineer - Special Projects Group
• Developed test infrastructure in Rust, Python, and C on bare-metal Linux systems for DSP/DAQ
• Architected low-level network drivers in Rust and debugging TCP/IP,UDP traffic with Wireshark and Lua
• Trained and tested AI/ML models utilizing PyTorch for an OpenCV, ffmpeg streaming applications
• Optimized data analysis and DSP jobs using CUDA for executing parallel computing on GPU in AWS EC2
• Synthesized actuator control systems via sysid testing, modeling, and frequency domain analysis in MATLAB
• Managed DevSecOps activities: OS upgrades, OpenVPN, PGP keygen, firewalls, CVE scans, TLS certs, YubiKeys
• Led efforts to establish scalable cloud CI/CD leveraging Git, Bash, Bazel, Jenkins, Docker, and Ansible
Jun 2021 - May 2024 (3yr, 0mo)
Cupertino, CA, USA

Pratt & Whitney Canada
Software Engineer - Test Facilities Computing
• Shipped production ECU test and build software with modern web stack: Rust, Node, Vue, TypeScript
• Designed embedded C drivers for RTOS control systems and high-speed ECU data acquisition and processing
• Maintained concurrent C++, C#, Rust microservices running on bare-metal CentOS, RHEL Linux servers
• Programmed TCP/UDP sockets for communication with APIs/SQL databases and debugged with Wireshark
• Introduced a new Agile work-flow with Git, Azure CI, Docker, and C++/C/ JavaScript unit testing
Aug 2019 - Jun 2021 (1yr, 11mo)
Remote

Tesla

Sep 2018 - Dec 2018 (0yr, 4mo)

Firmware Engineering (Co-op) - Energy Products

Palo Alto, CA, USA

- Coded MISRA compliant firmware in C for power electronic controls on embedded system's DSP's and MCU's
- Exposed to full-stack from RTOS kernel, serial drivers APIs (CAN, SPI), application-level controls and diagnostics
- Deployed an embedded self-test C framework on multiple ECUs eliminating manual debugging at EOL/field
- Employed a test-driven development mindset by writing C unit tests, SIL/HIL simulations,  regression
- Assured CI in an Agile environment with Atlassian tools,  Git,  Bash, code review/PR and  Jenkins builds

Apple

Aug 2017 - Aug 2018 (1yr, 0mo)

Controls Engineering (Co-op) - Special Projects Group

Cupertino, CA, USA

- Developed a hardware-in-the-loop system for validation of power electronic control algorithms in C on MCU
- Emulated and optimized high-fidelity discrete plant models on 32-bit Xilinx FPGA for low latency second control
- Deployed LabVIEW HMI for deterministic communication between PC, PXIe RTOS controller and FPGA
- Flashed microcontroller via JTAG, serial and Ethernet with the latest software builds for bring-up of PCB
- Applied DSP theory to convert continuous Simulink filters to discrete firmware in C for data acquisition
- Designed system harness to interface HIL and PCB from electrical schematics and NI hardware datasheets

Altaeros

Jan 2017 - Apr 2017 (0yr, 3mo)

Systems Engineering (Co-op) - R & D

Boston, MA, USA

- Performed numerical analysis in  Python on prototype of an autonomous aerostat's electromechanical system
- Utilized electronic lab equipment and LabVIEW HMI to log test data and analyze with 

Ontario Die International

May 2016 - Aug 2016 (0yr, 4mo)

Mechanical Engineering (Co-op) - R & D

Waterloo, ON, CAN

- Designed robotic components (electrical, hydraulic) of PLC/CNC bending systems in SOLIDWORKS
- Improved existing technology by applying lean, DFM and DFA principles to prototyping and research projects
- Automated tedious SOLIDWORKS tasks in VBA and C++ with the API in MS Visual Studio IDE
- Performed hands-on Q&A HMI testing, machined components, fabricated assemblies with power/hand tools

Pratt & Whitney Canada

Sep 2015 - Dec 2015 (0yr, 4mo)

Project Management (Co-op) - Turbofan Operations

Mississauga, ON, CAN

- Communicated with the OEM in French to assure delivery of a quality engine while exceeding expectations
- Developed Excel VBA programs allowing for improvements in methods of business metric preparation
- Collaborated with a multi-disciplinary team to develop logistics plans for new design incorporation

Linamar

Jan 2015 - Apr 2015 (0yr, 3mo)

Manufacturing Engineering (Co-op) - Skyjack

Guelph, ON, CAN

- Worked with a team of engineers to troubleshoot production issues at an aerial work platform manufacturer
- Increased process efficiency by organizing assembly stations and designing of jigs/fixtures with SOLIDWORKS

Projects

Golf Simulator

Mar 2026

Personal

- Deployed a WASM based golf ball physics simulation built with Bevy/

Sailboat Weather Station

Jun 2025

Personal

- Implemented  Python serial drivers for wind sensors and I2C IMU, integrated with 

Electric Drum Kit Trainer

Sep 2023

Personal

- Developed a  Rust based real-time MIDI striking pattern monitor on an NXP ARM running FreeRTOS

Robot Arm Controller

Apr 2019

ECE 488: Multi-Variable Controls

- Modeled and controlled MIMO non-linear system in  MATLAB using optimal LGC control methods

Heated Press System

Jan 2019

ME 482: Capstone Design Project

- Led electrical system efforts including harnessing/debugging and temperature/motor controls in C on MCU

Interests

Golf, off-road vehicles, sailing, hockey, tinkering, electronics, machine learning, cybersecurity, socializing (French, English)